Data Flow Diagram & User Stories

Date	19 September 2026
Team ID	
Project Name	FreelanceHub — MERN Stack Freelancing Platform
Maximum Marks	4 Marks

1. DFD Level 0: Context Diagram

The Context Diagram shows FreelanceHub as a single process, with the two external entities — the Client and the Freelancer — and the data that flows between them and the system.

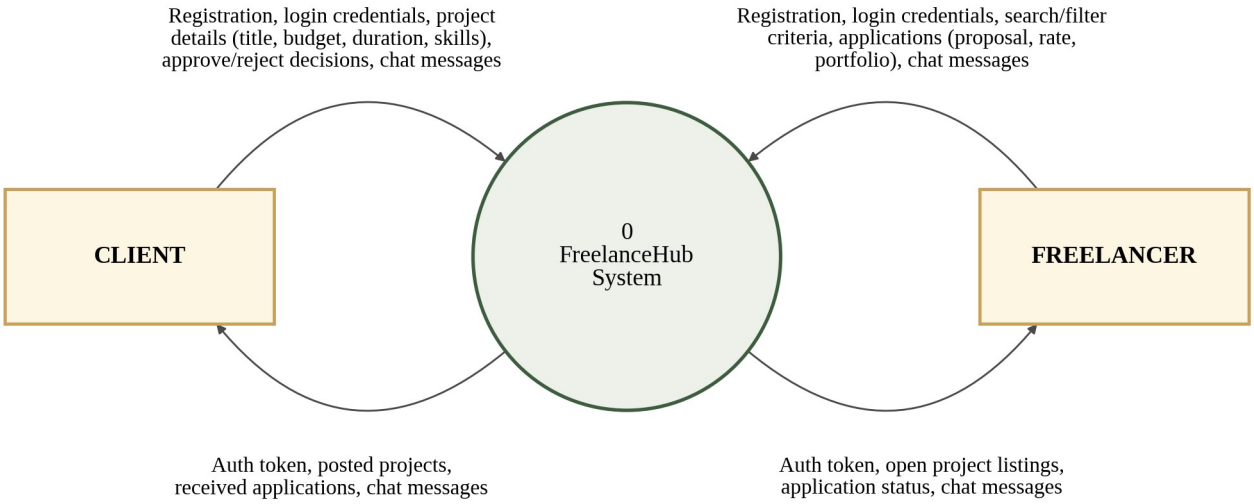


Figure 1: FreelanceHub Level 0 context diagram

2. DFD Level 1: Process Breakdown

The Level 1 diagram breaks the single system process into its six core processes and five data stores, showing exactly how data moves between the client, the freelancer, and MongoDB.

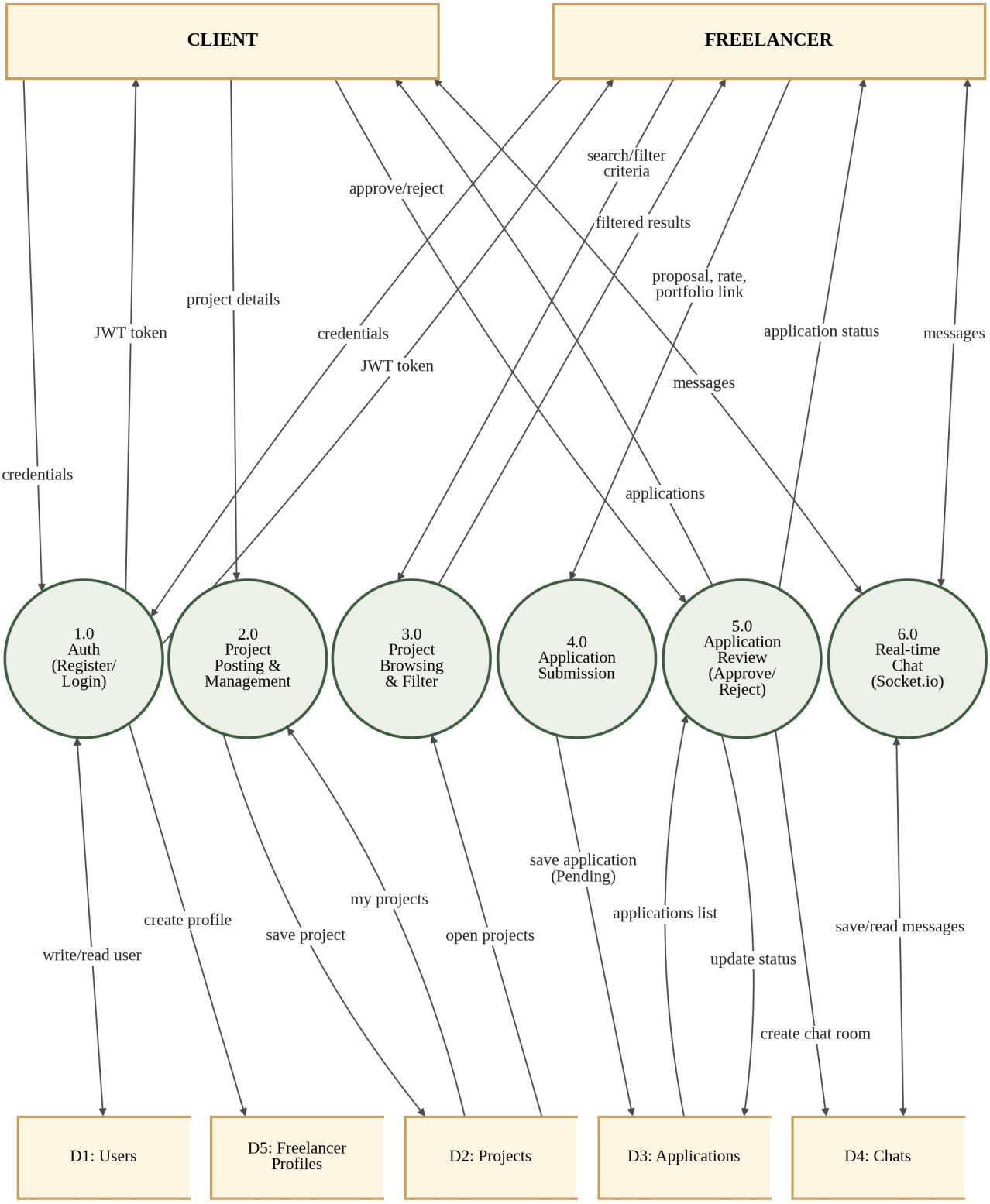


Figure 2: FreelanceHub Level 1 data flow diagram

3. Data Flow Diagram Entity and Process Dictionary

Element Identifier	Typology	Description & System Interactivity	Technology Mapping
Client	External Entity	Registers, logs in, posts projects, reviews applications, approves or rejects freelancers, and chats with approved freelancers.	React Frontend (SPA)
Freelancer	External Entity	Registers, logs in, browses and filters projects, submits applications, tracks application status, and chats with clients.	React Frontend (SPA)
Process 1.0 — Auth (Register/Login)	System Process	Handles registration and login; hashes passwords with bcrypt and issues signed JWTs with the user's role.	Express + bcrypt + JWT
Process 2.0 — Project Posting & Management	System Process	Lets clients create projects and view the projects they have posted.	Client-protected Express route + Mongoose
Process 3.0 — Project Browsing & Filter	System Process	Serves the list of open projects and filters them by skills and budget.	Express route + Mongoose query
Process 4.0 — Application Submission	System Process	Accepts a freelancer's proposal, rate and portfolio link and saves it with Pending status.	Freelancer-protected Express route
Process 5.0 — Application Review	System Process	Lists applications for a client's project, updates status to Approved or Rejected, and opens a chat room on approval.	Client-protected Express controller
Process 6.0 — Real-time Chat	System Process	Delivers messages instantly between the approved client and freelancer and stores chat history.	Socket.io + Express + Mongoose
D1: Users	Data Store	Stores name, email, hashed password, and role (client / freelancer / admin).	MongoDB collection (users)
D2: Projects	Data Store	Stores project title, description, budget, duration, required skills, status, and client reference.	MongoDB collection (projects)
D3: Applications	Data Store	Stores project and freelancer references, proposal, rate, portfolio link, and status.	MongoDB collection (applications)
D4: Chats	Data Store	Stores chat participants and messages with sender, text, and timestamp.	MongoDB collection (chats)
D5: Freelancer Profiles	Data Store	Stores freelancer profile details such as skills and portfolio.	MongoDB collection (freelancers)

4. User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Client / Freelancer	Registration	USN-1	As a user, I can register by entering my name, email, password, and choosing my role as Client or Freelancer.	Account is created and password is stored hashed.	High	Sprint-1
Client / Freelancer	Login	USN-2	As a user, I can log in with my email and password and receive a secure JWT token.	I am redirected to my role-based dashboard.	High	Sprint-1
Client / Freelancer	Role-based Access	USN-3	As a user, I can only access pages and actions allowed for my role.	Restricted pages redirect or return "Access denied".	High	Sprint-1
Client	Project Management	USN-4	As a client, I can post a project with title, description, budget, duration, and required skills.	Project is saved and visible to freelancers.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Client	Project Management	USN-5	As a client, I can view and manage all the projects I have posted.	Only my own projects are listed on my dashboard.	Medium	Sprint-2
Freelancer	Project Browsing	USN-6	As a freelancer, I can browse and filter open projects by skills and budget.	Matching open projects are displayed.	High	Sprint-2
Freelancer	Applications	USN-7	As a freelancer, I can apply to a project with a proposal, rate, and portfolio link.	Application is saved with Pending status.	High	Sprint-2
Client	Application Management	USN-8	As a client, I can view all applications received for each of my projects.	All applications for the project are listed with details.	High	Sprint-2
Client	Application Management	USN-9	As a client, I can approve or reject a freelancer's application.	Status updates and chat opens on approval.	High	Sprint-3
Freelancer	Application Tracking	USN-10	As a freelancer, I can track the status of my applications.	Status shows Pending, Approved, or Rejected.	Medium	Sprint-3
Client / Freelancer	Real-time Chat	USN-11	As an approved client or freelancer, I can chat with the other party in real time.	Messages appear instantly and are saved in the database.	High	Sprint-3
Client / Freelancer	Notifications	USN-12	As a user, I receive toast notifications for successful actions and errors.	A success or error message appears after each action.	Medium	Sprint-4
Client / Freelancer	UI/UX	USN-13	As a user, I want a consistent, responsive interface across every page.	Pages display correctly on desktop and mobile.	Low	Sprint-4